

ENGINEERING INSPECTION REPORT

Physics-Guided Bearing Fault Diagnosis

Generated: July 16, 2026 at 07:31

1. Executive Summary

Parameter	Value
Detected Fault	Inner Race Fault
Model Confidence	84.3%
Health Index	0.2%
Physics Confidence	100.0%
System Status	CRITICAL
Recommended Action	Replace Bearing Immediately

2. Bearing Kinematics & Fault Frequencies

The following fault characteristic frequencies were computed from bearing geometry and shaft speed using ISO bearing kinematics formulas.

Frequency	Symbol	Value (Hz)	Description
Ball Pass Frequency Outer Race	BPFO	113.94	Fault indicator frequency
Ball Pass Frequency Inner Race	BPFI	163.56	Fault indicator frequency
Ball Spin Frequency	BSF	83.45	Fault indicator frequency
Fundamental Train Frequency	FTF	12.66	Fault indicator frequency
Shaft Rotation Frequency	shaft_hz	30.83	Fault indicator frequency

3. Model Performance Metrics

Metric	Value	Benchmark
Accuracy	100.00%	≥ 95%
Precision	100.00%	≥ 95%
Recall	100.00%	≥ 95%
F1 Score	100.00%	≥ 95%

4. Physics Validation

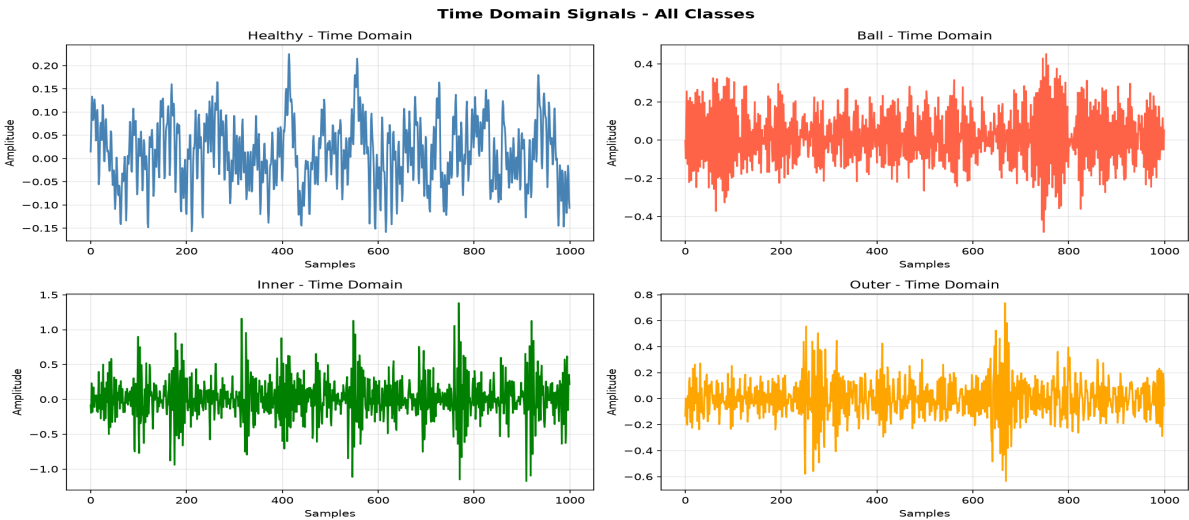
The AI prediction was cross-validated against physics-based bearing kinematics. Energy was analysed at fault characteristic frequencies, harmonics, and sidebands.

Physics Parameter	Result
Target Frequency	163.56 Hz
Physics Confidence	100.00%

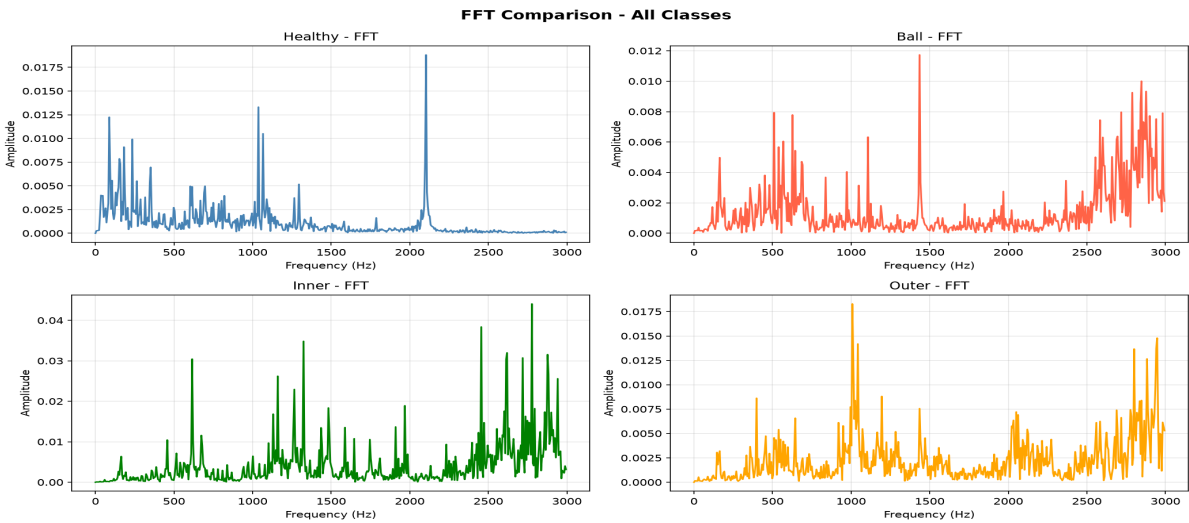
Validation Message	Physics confidence: 100.0%
Harmonic Analysis	BPFO × 1,2,3 BPFI × 1,2,3 BSF × 1,2,3
Sideband Analysis	Sidebands at fault_freq ± k × shaft_hz

5. Signal Analysis Plots

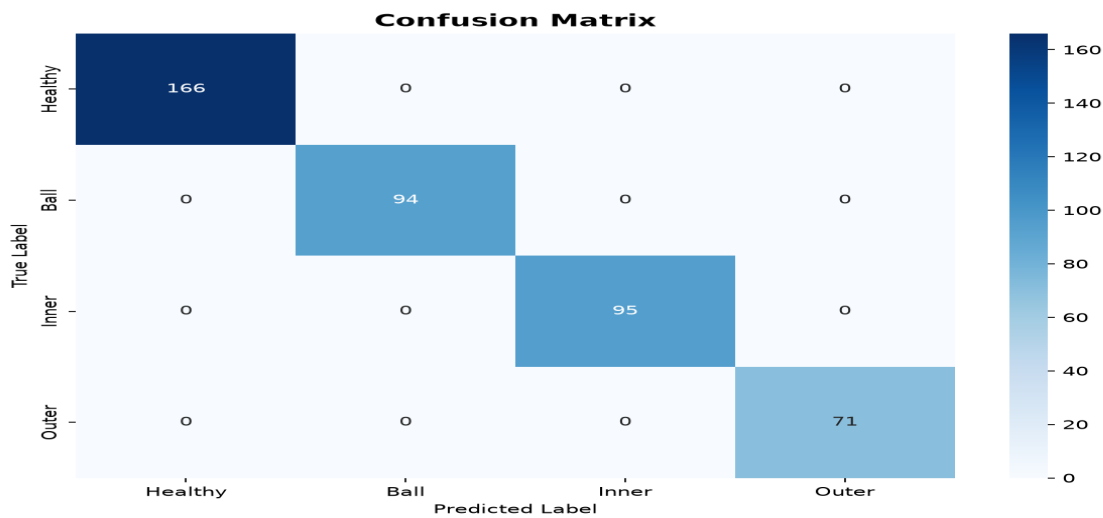
Time Domain Signals - All Classes



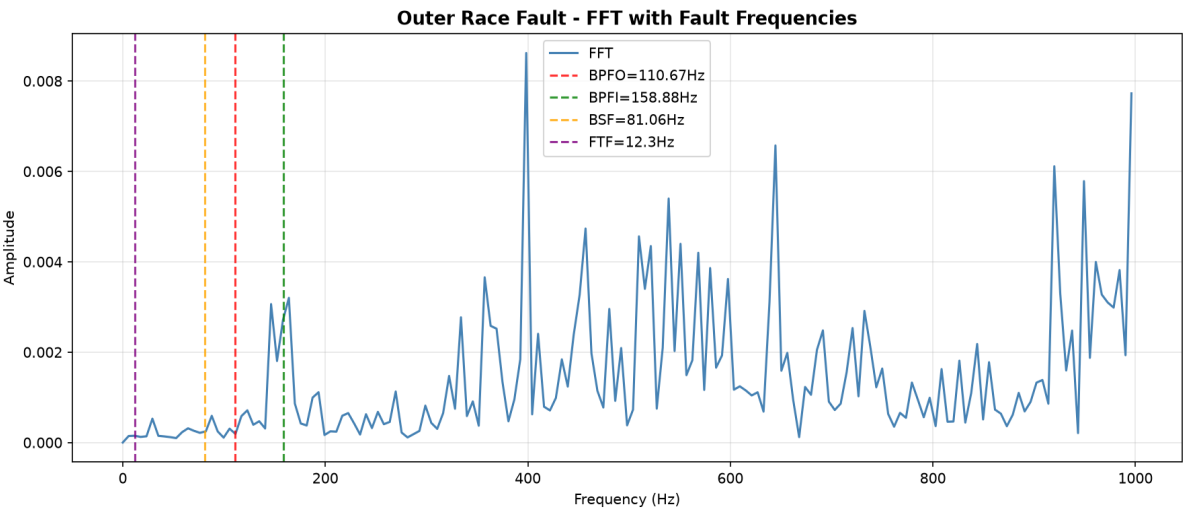
FFT Comparison - All Classes



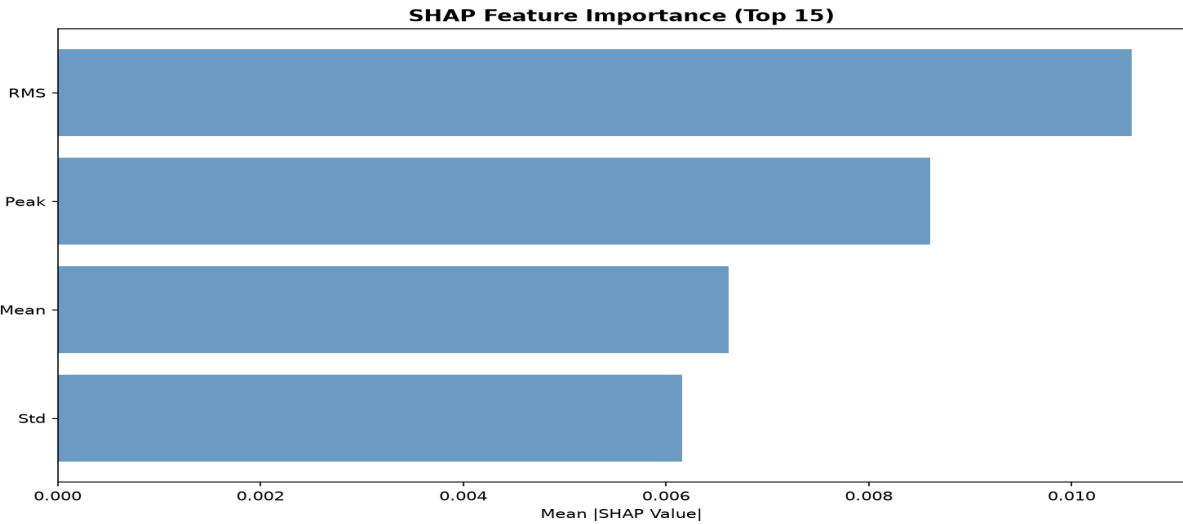
Confusion Matrix



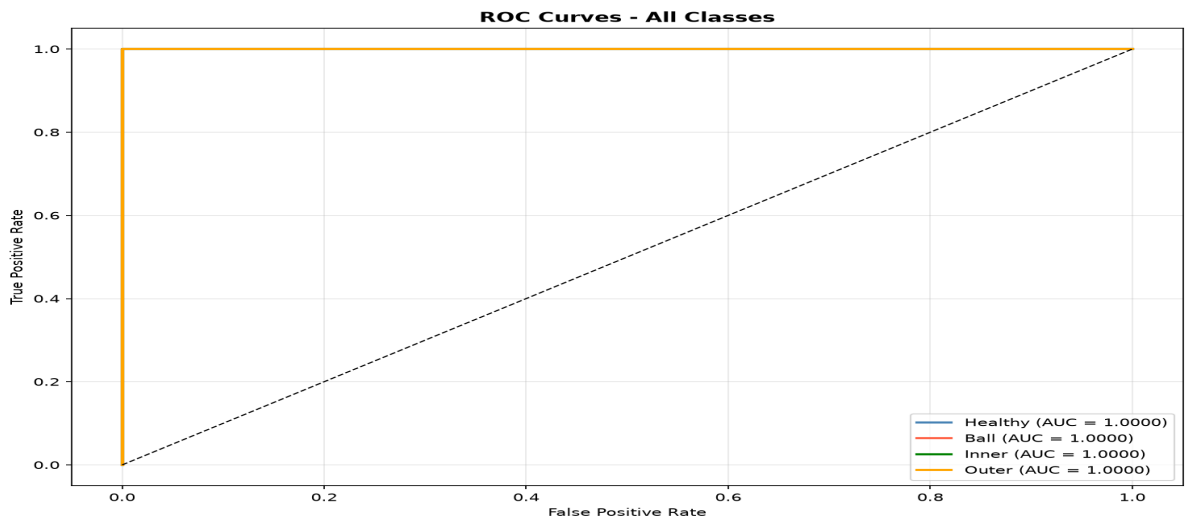
FFT with Fault Frequencies



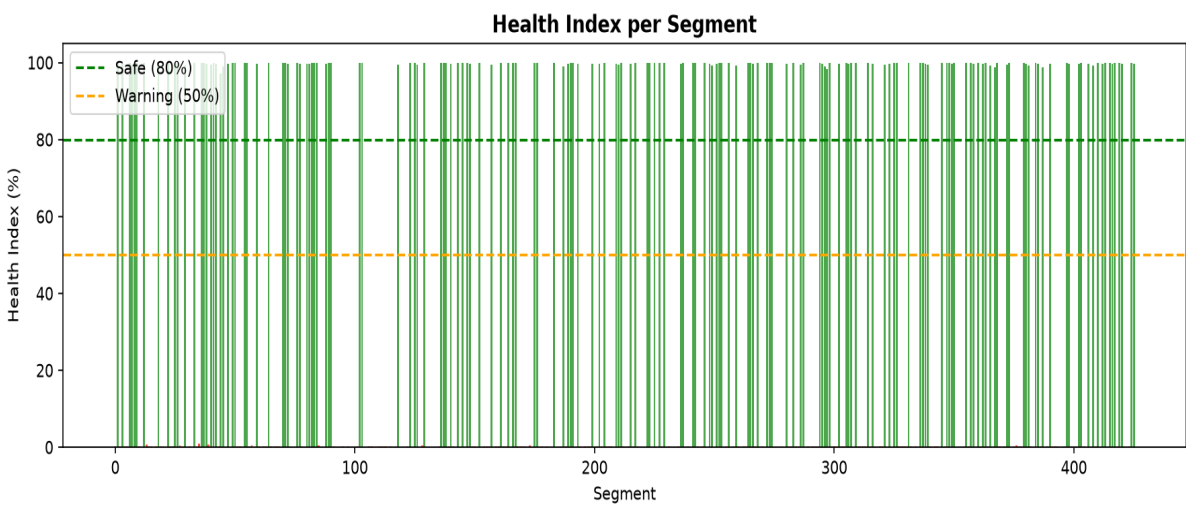
SHAP Feature Importance



ROC Curves - All Classes



Health Index per Segment



6. Top Extracted Features

47 features extracted across statistical, frequency, wavelet and physics domains.

#	Feature Name	Value	Domain
0	RMS	0.171217	Statistical
1	Peak	1.257594	Statistical
2	Std	0.171217	Statistical
3	Mean	-0.000000	Statistical
4	Kurtosis	12.964672	Statistical
5	Skewness	-0.067006	Statistical
6	Crest_Factor	7.345018	Statistical
7	Peak_to_Peak	2.469176	Statistical
8	Impulse_Factor	11.206445	Statistical
9	Shape_Factor	1.525720	Statistical
10	Margin_Factor	14.213449	Statistical
11	Dominant_Freq	1388.671875	Frequency
12	Spectral_Centroid	993.842573	Frequency
13	Spectral_Energy	0.002420	Frequency
14	Spectral_Entropy	5.228826	Frequency
15	Low_Band_Ratio	0.006364	Frequency
16	Mid_Band_Ratio	0.188748	Frequency
17	High_Band_Ratio	0.804887	Frequency
18	Peak_Amplitude	0.033111	Frequency
19	Wavelet_L0_Energy	1.903187	Wavelet

7. Maintenance Recommendation

Field	Details
Fault Detected	Inner Race Fault
Health Index	0.2%
Status	CRITICAL
Recommended Action	Replace Bearing Immediately
Next Inspection	Within 24 hours
Engineer Sign-off	
Date	2026-07-16

This report was automatically generated by the Physics-Aware Predictive Maintenance System. Dataset: CWRU Bearing Dataset (Case Western Reserve University). Models: Random Forest, SVM, XGBoost. Physics Engine: Bearing kinematics + harmonic + sideband analysis.